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INCAR^2
Interdisciplinary Center
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- Applied Research

Are fishing decisions flexible?

Participation, target, and landing choices in the U.S. West Coast pelagic fishery

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Motivation — the human margin of climate adaptation

- Climate **reallocates** forage-species distributions — the 2014–2016 heatwave shifted squid **northward**.
- To keep fishing, harvesters can **follow** species, **switch** species, **switch ports**, halt, or exit.
- Standard projections **fix behavior** — missing the margin where fishers actually *adapt*.

Question

How did **availability, prices, costs, weather** shape **daily** decisions to participate, target, and land — over 2013–2017, spanning the **2015 sardine closure** and the **2014–2016 heatwave**?

> We measure **demonstrated flexibility** across the joint choice *participate* × *target* × *land*.

The key input: availability from SDMs, not past catch

Past catch (the usual proxy)

- ✗ **Endogenous** — reflects where vessels already *chose* to fish.
- ✗ **Selection bias** — only observed at fished sites (Chen et al., 2023).
- ✗ **Gaps** — undefined on days/ports with no fishing.

SDM habitat index (ours)

- ✓ **Exogenous** environmental input to the choice.
- ✓ **No gaps** — an availability value for *every* port–day.
- ✓ **Time-varying** — captures environmentally-driven interannual shifts.

An **exogenous**, gap-free, climate-responsive proxy $\Rightarrow$ cleaner identification *and* climate-ready counterfactuals.

Setting — four fleet segments, four portfolios

U.S. West Coast CPS fishery: squid, sardine, anchovy, mackerels (+ non-CPS crab, salmon, tuna); purse seine. Segments from cluster analysis (Quezada et al., 2024).

Fleet segment	Main target(s)	Trips N	Vessels/yr
S. CCS industrial squid specialist	Market squid	29,160	42
Roving squid–sardine generalist	Squid & sardine	6,806	14
PNW sardine specialist	Pacific sardine	2,581	6
S. CCS forage-fish diverse	Diverse CPS	2,481	5

> Same shock, **different portfolios** — the unit of adaptation is the *fleet segment*.

The model — nested logit over *participate* × *target* × *port*

Utility of alternative $j = (p, s, l)$ for vessel i on day t:

$$U_{itj} = \delta_j + \beta_{1,s}\,\mathrm{Avail}_{tj} + \beta_{2,g}\,\mathrm{Price}_{tj} + \ldots + \varepsilon_{itj}$$

- **Availability** = daily **SDM** index (species radius); CPUE index otherwise.
- Also: price, distances (catch, center of gravity), wind, closures, unemployment.
- **State dependence** (chosen in last 30 days) — proxy for processor ties / port capacity.

Nesting (by segment)

- Correlation within **participation**, then within **species**...
- ...except **forage-diverse**: nests by **port**.
- Crab/salmon split off (costly gear switch).

Apollo (R). Dominant species = target (**93.6%** of trip revenue); ∼60/40 no-participation/trips.

Availability enters at **daily** frequency and is **exogenous** to the vessel's choice.

Descriptives — persistence *and* frequent switching

	Species		Ports	
Fleet segment	Switches	Unique	Switches	Unique
S. CCS industrial squid specialist	**19.8**	2.6	**20.7**	3.3
Roving squid–sardine generalist	4.6	1.7	9.5	2.7
PNW sardine specialist	3.8	2.4	4.7	2.1
S. CCS forage-fish diverse	10.8	2.4	5.1	2.0

Vessel-level means, 2013–2017. Squid specialist: **switch rate 0.36**, **median run 2 days** — persistence *and* regular switching.

Short-run **persistence** + frequent **switching** ⇒ recent choices matter (motivates state dependence).

What drives the choice? Availability signs by segment

Availability of...	Squid spec.	Roving gen.	PNW sardine	Forage diverse
Market squid	+2.45***	+1.75***	—	+1.61*
Northern anchovy	+1.35***	+4.19***	+1.16***	**−1.88*****
Pacific sardine	−1.78**	0.23	−0.59**	0.23
Chub & jack mackerel	−3.94***	−1.48	0.87	−1.92***
Non-CPS (crab/salmon)	—	+5.74***	+10.3***	—
Expected price (CPS/tuna)	0.26	+4.60***	+3.21***	+0.72***
State dependence (30-day)	+2.63***	+2.86***	+2.26***	+3.09***

Wind, distance-to-catch, distance-to-CG all $(-)$; unemployment $(+)$. Negative anchovy for forage-diverse: anchovy–squid availability correlate $r = -0.29$.

State dependence is **large** (coef. $\approx$2.3–3.1), yet availability and price **stay significant** $\Rightarrow$ flexibility is **genuine**, not just habit.

Two adaptation modes — “on the move” vs. “in place”

Adapt *on the move* (species nests)

- Low dissimilarity $\lambda \Rightarrow$ strong within-species correlation across **ports**.
- Anchovy $\lambda = 0.49$ (squid spec.); sardine $\lambda = 0.40$ (roving, PNW).
- $\Rightarrow$ **follow the species across ports**.

Adapt *in place* (port nests)

- Forage-diverse nests by **port**; Santa Barbara $\lambda = 0.49$ vs. LA/Monterey $\lambda = 0.73$.
- Substitution across **species within a port**.
- $\Rightarrow$ **switch species, stay put** (by relative price).

The adaptation *mode* is a structural property of the fleet (sensu Samhouri et al., 2024).

Elasticities — price drives *entry*, availability drives *targeting*

Extensive margin (participate)

- Price dominates for mobile fleets: squid **1.71** (roving); anchovy **0.43** (PNW).
- Non-CPS availability matters where crab/salmon are relevant: crab **0.24**, sockeye **0.19** (PNW).

Intensive margin (conditional target)

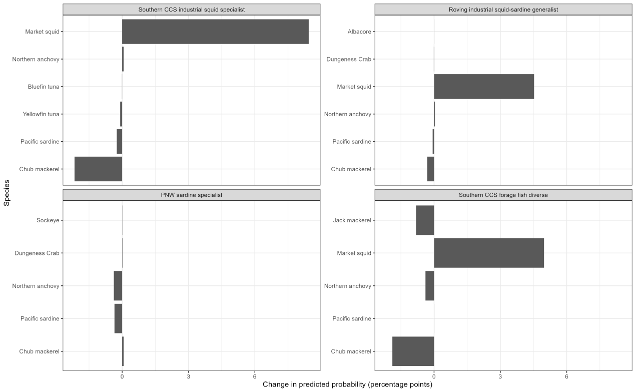
- Availability often > price: anchovy **2.55**, crab **2.12** (roving); crab **3.85**, sockeye **3.71** (PNW).
- Several > 1 ⇒ more-than-proportional responses.
- Exception: forage-diverse is **price**-driven.

> **Prices** move participation; **availability** moves which species, once at sea.

Scenario — redistribute squid availability southward

Replace squid availability with a later observed window (SDM centroid **42.99°N → 40.66°N**); hold prices, closures, distances, state fixed.

- Squid spec.: ↑ **squid**, ↓ chub, ↓ no-participation.
- Roving: ↑ **squid**. PNW: small (no squid target).
- Forage-diverse: large reallocations.



Δ predicted probability (pp) by target species, per segment.

> Species availability is a **key short-run driver** — with **heterogeneous** responses by segment.

Takeaways

1. Harvesters are **flexible**: participation, target, and landing respond to **availability, prices, costs, weather — even controlling for strong state dependence**, so it is not just habit.
2. **Two adaptation modes** — "on the move" (substitute across *ports*) vs. "in place" (switch *species* within a port).
3. **SDM-based** availability is exogenous and gap-free; OOS pseudo-ρ^2 **0.22–0.44** $\Rightarrow$ a reliable, climate-ready policy tool.

> Adaptive capacity is a property of the **bundle** (portfolio $\times$ ports $\times$ institutions $\times$ market), not of fishers alone.

Thank you! / ¿Gracias?

Questions?

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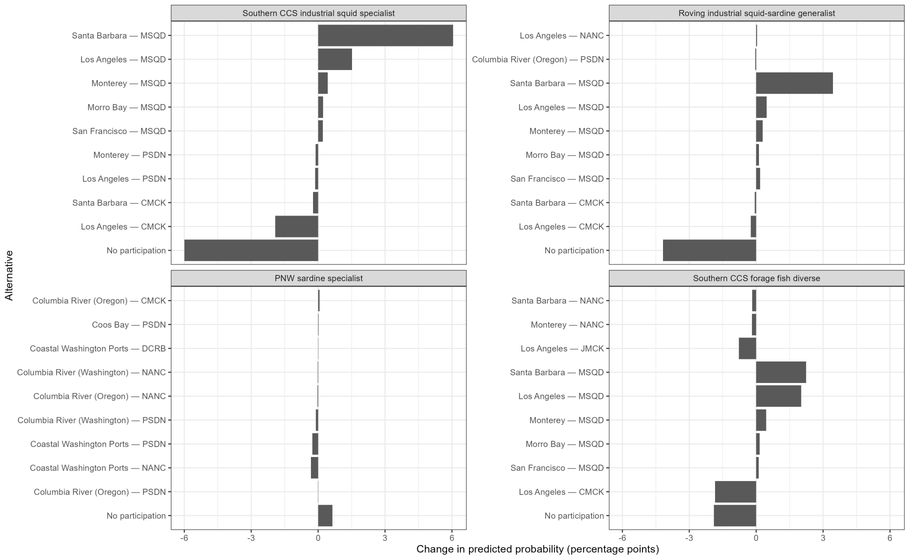
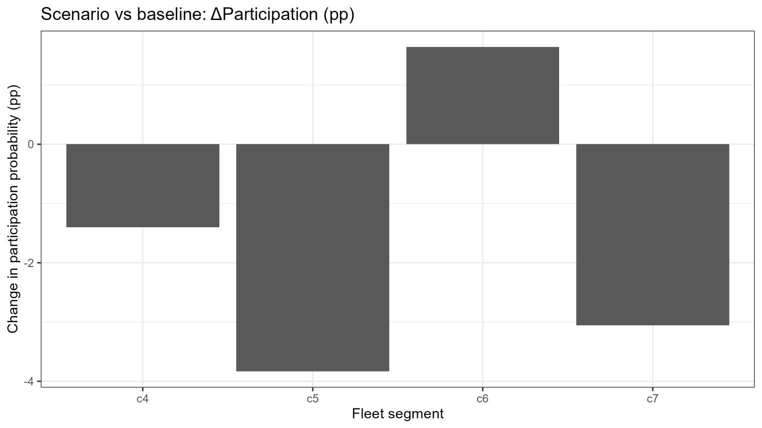
Backup

Backup • Scenario — change by port × species alternative



Backup • Scenario — change in participation



Δ probability of no-participation under the redistribution scenario, by segment.

Backup • Dissimilarity parameters (λ)

Nest	Squid spec.	Roving gen.	PNW sardine	Forage diverse
Participation (CPS/Tuna)	0.977	0.752	0.532	0.730
Market squid	0.777	0.752	—	—
Pacific sardine	0.617	**0.403**	**0.404**	—
Northern anchovy	**0.494**	—	0.532	—
Santa Barbara (port)	—	—	—	**0.490**
Los Angeles / Monterey (port)	—	—	—	0.730

Lower $\lambda =$ stronger within-nest correlation (more substitution within the nest).

Backup • Variable construction

- **Availability:** binomial GAM SDMs (Muhling et al., 2019, 2020); mean presence within a species radius (60 km sardine/anchovy/mackerel, 90 km squid). CPUE z-score $\rightarrow$ sigmoid for non-SDM species.
- **Price:** predicted from $\ln(\text{price}) = \alpha + T_y + \gamma_m + \gamma_p$ (year trend, month & port FE); robustness with 30-day moving average.
- **Costs:** Haversine distance port$\rightarrow$catch area and port$\rightarrow$center of gravity.
- **Weather:** max ERA5 surface wind within 220 km. **Opportunity cost:** state unemployment.

Backup • Limitations

- **Aggregation bias:** port-area resolution may mask finer spatial heterogeneity (Dépalle et al., 2021).
- **Choice-set truncation:** rare alternatives folded into the outside option ($\geq$0.5% filter); IIA assumed for omitted alternatives.
- **Price endogeneity:** predicted / MA prices reduce but do not eliminate simultaneity $\Rightarrow$ interpret price coefficients as **reduced-form** behavioral responses.
- **SDM proxy:** habitat suitability $\neq$ catch rate exactly $\Rightarrow$ availability effect may be attenuated.

> Main qualitative result — **availability drives targeting** — is robust across price specifications.